ENGINE COOLANT TEST

AFFECTED VEHICLES: CERTAIN 2001 MODEL YEAR TAURUS/SABLE

OVERVIEW

The affected vehicles may have been built with engine coolant having a higher than recommended (50%) ethylene glycol concentration in the cooling system. **The coolant test that will be performed is to determine the ethylene glycol concentration, not a simple freeze protection test. Therefore, the required tool is the Rotunda Coolant Tester 164-R1025 (also commercially available as the Thexton 106 coolant tester).** This tester was supplied for a previous program and is capable of determining an improper concentration of ethylene glycol in the coolant.

- On vehicles equipped with Duratec (4v) engines that have a high ethylene glycol concentration, a special drain and fill procedure has been developed which will reduce the ethylene glycol concentration to 50%.
- On vehicles equipped with Vulcan (2v) engines that have a high ethylene glycol concentration, an engine load test and a check for Diagnostic Trouble Codes (DTCs) will be performed. If any DTCs are present or the vehicle exhibits a misfire condition, a spark plug inspection and compression test will be required to determine if internal engine damage has occurred. If damage is found, the engine must be replaced. Refer to the procedure in attachment II to obtain claiming and engine ordering instructions. If no damage is found, you will reinstall the spark plugs and perform the special drain and fill procedure developed to reduce the ethylene glycol concentration to 50%.

ENGINE COOLANT TEST

NOTE: Allow the engine to cool before performing this test.

1. **WARNING: TO AVOID PERSONAL INJURY, DO NOT UNSCREW THE COOLANT PRESSURE RELIEF CAP WHILE ENGINE IS OPERATING OR HOT. THE COOLING SYSTEM IS UNDER PRESSURE; STEAM AND HOT LIQUID CAN COME OUT FORCEFULLY WHEN THE CAP IS LOOSENED SLIGHTLY.**

   Wrap a shop cloth around the coolant pressure relief cap on the coolant reservoir. Slowly turn the cap counterclockwise to release the cooling system pressure.

2. When all the pressure has been released, remove the coolant reservoir cap.

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3. Insert the Rotunda Coolant Tester 164-R1025 into the coolant reservoir. Fill the tester to the bottom of the bulb as shown. See Figure 1.
   A. If four (4) or five (5) discs float, no further service is required. Return the vehicle to the owner. See Figure 2.
   B. If all six (6) discs float, this indicates an "IMPROPER" ethylene glycol concentration. See Figure 2.
   - For Duratec (4v) engines, proceed to the COOLANT DRAIN AND FILL section of this procedure.
   - For Vulcan (2v) engines, proceed to the ENGINE TESTING section of this procedure.

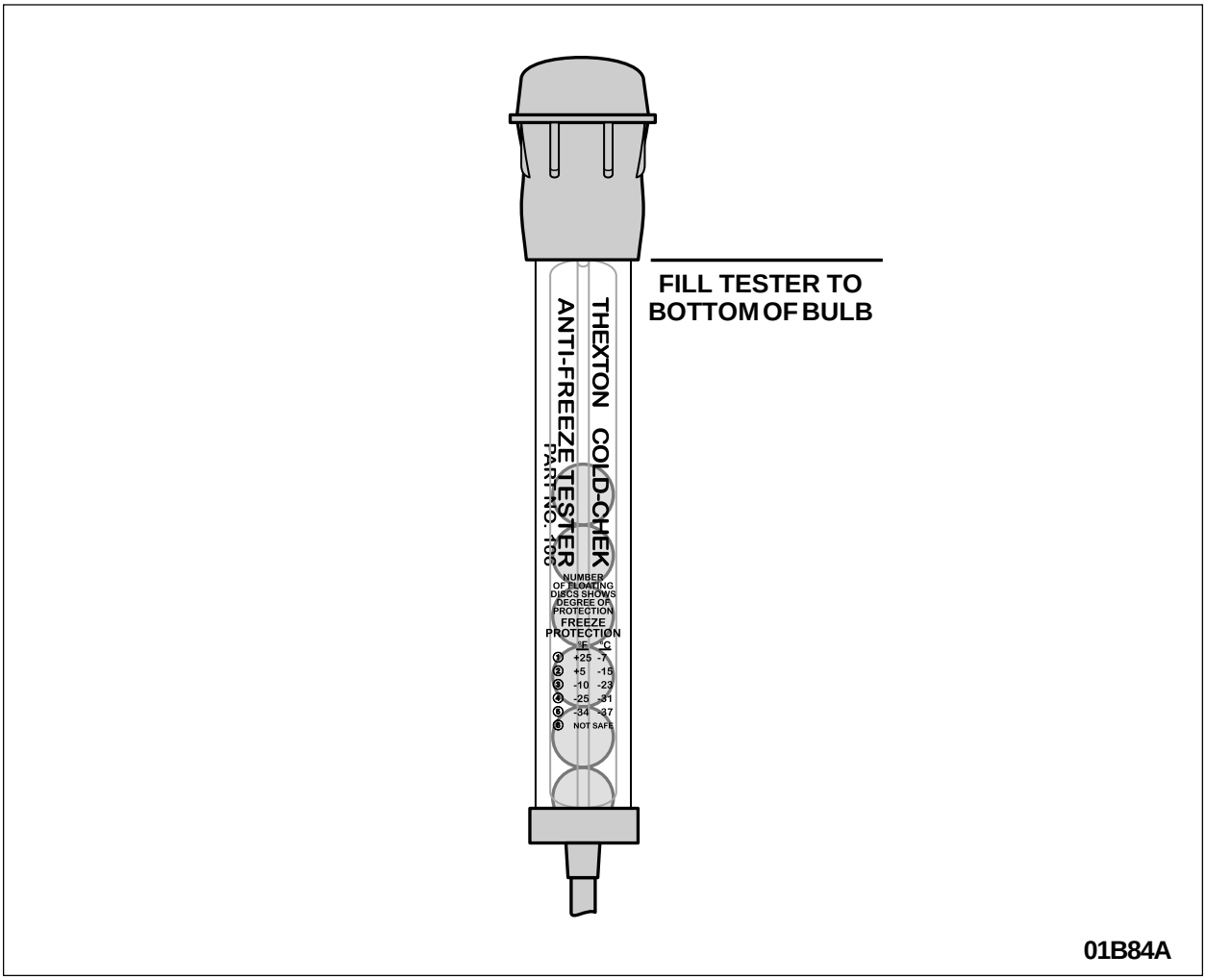


FIGURE 1

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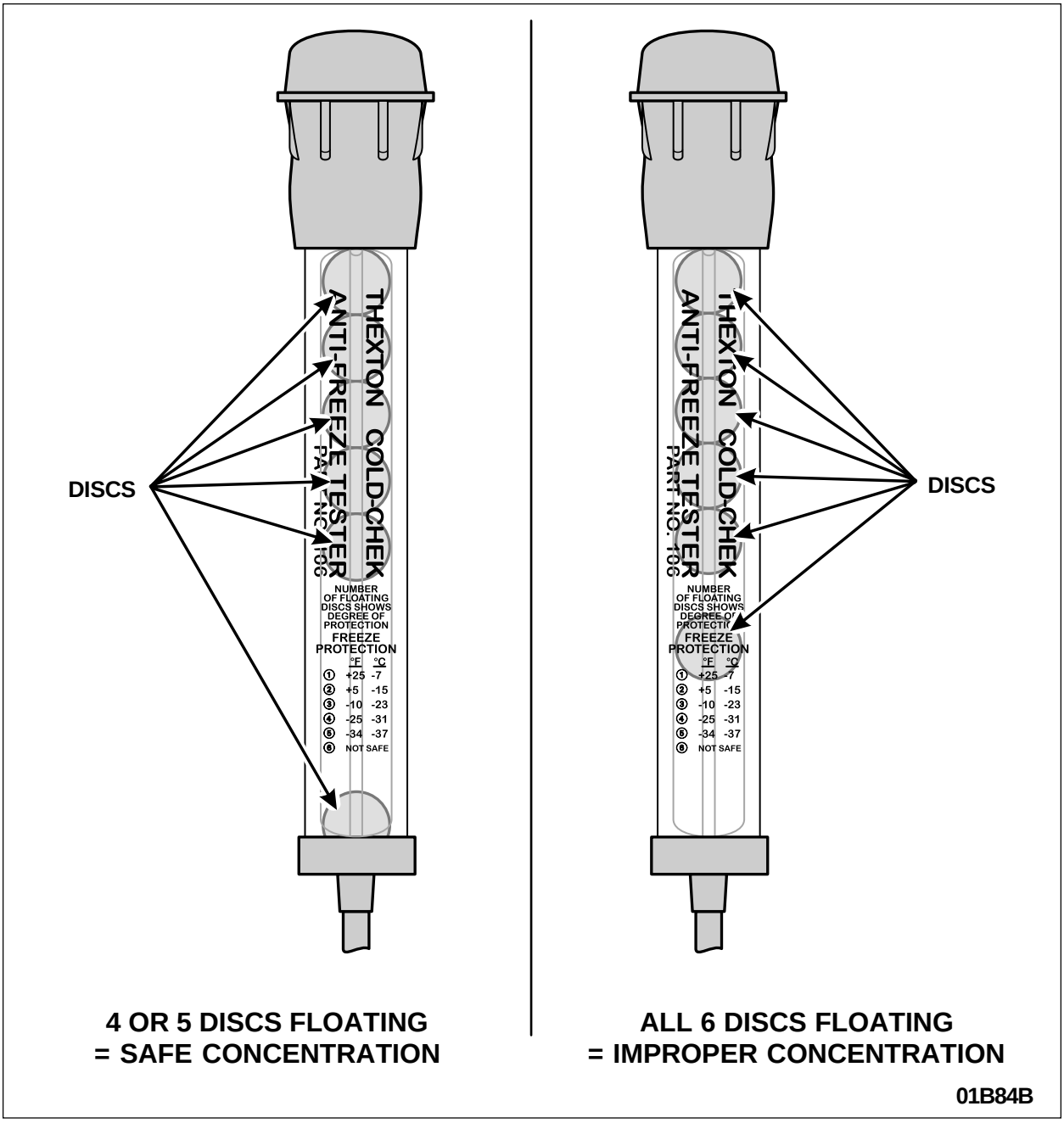


FIGURE 2

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ENGINE TESTING *(VULCAN 2v ENGINES ONLY)*

NOTE: The following steps should only be performed on vehicles equipped with Vulcan (2v) engines, that have an improper concentration of ethylene glycol in the coolant system.

1. **CAUTION: This test must be performed in a safe area, clear of all building structures and personnel. The parking brake must be applied firmly while carrying out the engine load test.**

   Perform the following engine load to determine if a misfire condition exists in the engine.
   - Apply the parking brake firmly.
   - Start the engine.
   - Depress the brake pedal and place the transmission in DRIVE.
   - Accelerate the engine to 2,000 rpm and hold it for 10 seconds. Any misfire in the engine should be evident with the engine under a load.
   - Place the transmission in NEUTRAL and run the engine at 1,000 rpm for about 15 seconds to cool the torque converter.
   - Place the transmission in PARK and turn off the engine. Release the parking brake.
2. Using the New Generation Star (NGS) Tester or the Worldwide Diagnostic System (WDS), check for any engine performance-related diagnostic trouble codes (DTCs).
   - If no DTCs are present AND the vehicle did not exhibit any misfire condition during the engine load test, proceed to the COOLANT DRAIN AND FILL section of this procedure.
   - If there are DTCs present OR the vehicle exhibited a misfire condition during the engine load test, clear the DTCs (if necessary) and proceed to ENGINE INSPECTION.

ENGINE INSPECTION *(VULCAN 2v ENGINES ONLY)*

NOTE: The engine must be at operating temperature before starting this inspection. Do not allow the engine to cool before performing the compression test on the engine.

1. **CAUTION: The spark plugs MUST be marked or kept in order so they can be reinstalled into their original cylinder.**

   Remove all of the spark plugs from the engine. Inspect the spark plugs for damage due to pre-ignition or overheating, such as damaged insulators or burnt (or melted) electrodes.
   - If any spark plug(s) show damage, a ***new*** engine must be installed. However, you MUST first complete this engine inspection and record the results. The engine cylinder compression readings will be required for ordering the ***new*** engine.
2. Install a battery charger on the vehicle.
3. Prepare the engine for the compression test by setting the throttle plate in the wide-open position.
4. Install a compression tester in the No. 1 cylinder.
5. Install an auxiliary starter switch in the starting circuit.
6. With the ignition switch in the OFF position, and using the auxiliary starter switch, crank the engine a minimum of five compression strokes and record the highest reading. Note the approximate number of compression strokes required to obtain the highest reading.
7. Repeat the test on each cylinder, cranking the engine approximately the same number of compression strokes.

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8. Compare the highest and lowest cylinder compression readings. The lowest cylinder should be a minimum of 85 percent of the highest cylinder reading. For example, if the highest cylinder reading is 175 psi, the lowest cylinder reading should be a minimum of 149 psi (175 x 0.85).
   - If all of the cylinder readings are at least 85 percent of the highest reading **AND** there were no damaged spark plugs found during the inspection, install the spark plugs into their original cylinder positions. Tighten spark plugs to 15 Nm (11 lb-ft). Then, proceed to the COOLANT DRAIN AND FILL section of this procedure.
   - If any of the cylinder readings are less than 85 percent of the highest reading **OR** a damaged spark plug(s) was found during the inspection, internal engine damage has occurred. Install a ***new*** engine. Refer to the service procedures in Section 303-01A of the Workshop Manual for engine replacement procedure. Refer to the procedure in attachment II to obtain claiming and engine ordering instructions.

NOTE: If, after passing the spark plug inspection and compression test, the vehicle still exhibits any driveability concerns or a DTC persists, they are unrelated to this program. Further diagnostics should be covered under any remaining base warranty the vehicle has.

COOLANT DRAIN AND FILL *(ALL ENGINES)*

1. Remove the front splash shield to access the radiator draincock.
2. NOTE: Do not reuse any of the coolant removed from the vehicle. Dispose of all coolant properly.

   With the car level, open the radiator draincock and drain five (5) quarts of coolant from the radiator on Vulcan (2v) engines, or four (4) quarts on Duratec (4v) engines.
3. Close the radiator draincock.
4. NOTE: Due to the high concentration of ethylene glycol in the coolant remaining in the engine, refilling the cooling system with pure water will result in the recommended 50/50 coolant mixture.

   Fill the cooling system with pure water (approximately five [5] quarts) through the coolant reservoir until the level is between the “COOLANT FILL LEVEL” marks.
5. Select the maximum heater temperature and blower motor speed settings. Position the control to discharge air at A/C vents in the instrument panel.
6. Start the engine and allow to idle until normal operating temperature is reached. Hot air should discharge from A/C vents. The engine coolant temperature gauge should maintain a stabilized reading in the middle of the NORMAL range and the upper radiator hose should feel hot to the touch.
7. Install the coolant reservoir cap.
8. Allow the engine to run at operating temperature for a minimum of fifteen (15) minutes to allow the water and coolant to thoroughly mix.
9. Shut the engine off and allow to cool.

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10. **WARNING: TO AVOID PERSONAL INJURY, DO NOT UNSCREW THE COOLANT PRESSURE RELIEF CAP WHILE ENGINE IS OPERATING OR HOT. THE COOLING SYSTEM IS UNDER PRESSURE; STEAM AND HOT LIQUID CAN COME OUT FORCEFULLY WHEN THE CAP IS LOOSENED SLIGHTLY.**

    Wrap a shop cloth around the coolant pressure relief cap on the coolant reservoir. Slowly turn the cap counterclockwise to release the cooling system pressure.

11. When all the pressure has been released, remove the coolant reservoir cap.
12. Using the Rotunda Coolant Tester 164-R1025, test the coolant in the reservoir. Five (5) discs should be floating, indicating a 50/50 (recommended) coolant mixture. If less than five discs are floating, adjust the coolant concentration as required. Then, repeat steps 5 through 12 until the test indicates a 50/50 coolant mixture in the system.
13. Install the coolant reservoir cap.
14. Install the front splash shield.

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